

Hello everyone. This is Part 2 of BLA 2 in my AWS Machine Learning Engineer Associate MLA-C02 series. Part 1 explained how data can be collected, ingested, formatted, and stored.

In this presentation, I will discuss the purpose of transformation, the main AWS services that support data preparation, and common preprocessing methods. I will then explain feature engineering and SageMaker Feature Store. Because MLA-C02 includes generative AI, I will also cover embeddings, text preparation, and document preparation for Retrieval Augmented Generation, or RAG. The final topic is protecting sensitive information through masking, redaction, and anonymization.

Transformation connects raw information to model-ready input through a sequence of related steps. The first step is inspection.

Cleaning addresses known problems such as invalid entries, duplicate records, inconsistent units, and missing values.

The final step is validation.

AWS provides several services for data transformation, and each one serves a different type of workflow. AWS Glue is a managed data integration service. It can discover data, maintain metadata in the Glue Data Catalog, run extract-transform-load jobs, and move information across data sources. Glue is useful when preparation is part of a broader data engineering pipeline.

AWS Glue DataBrew provides visual data preparation. Users can profile datasets, identify quality issues, and build transformation recipes through an interface.

Preprocessing changes values so a model can interpret them more effectively. Scaling adjusts numeric values into a selected range. This can prevent a feature with large numbers from dominating another feature simply because of its units. For example, annual income and the number of purchases may have very different ranges even when both matter to the prediction.

Standardization centers values around their mean and describes them using standard deviation.

Encoding converts categorical information into a numerical representation. A model cannot directly interpret labels such as small, medium, and large unless the preparation process represents them appropriately.

Binning groups continuous values into ranges, such as age groups or spending bands. Log transformations can reduce strong positive skew and limit the influence of extremely large values.

A feature is an input that provides information to a model. Feature engineering creates or improves those inputs using knowledge of the data and the business problem. A timestamp might become hour of day, day of week, or time since a previous event. Two numeric fields might form a ratio. Transaction records might become a total number of purchases during the last thirty days.

Useful feature engineering makes important patterns easier to learn. Irrelevant, highly correlated, unstable, or leaked features can reduce performance.

SageMaker Feature Store can organize, store, share, and serve features. A centralized feature definition helps different teams reuse the same logic. It can also support consistency between offline features used for training and online features used for low-latency inference.

An embedding converts text, images, or other content into a numerical vector. The numbers do not represent a simple list of keywords. They represent patterns learned by an embedding model. Items with similar meaning are expected to appear closer together in the vector space, even when they do not use exactly the same words.

Embeddings support semantic search, recommendations, clustering, duplicate detection, and Retrieval Augmented Generation. In a RAG system, documents are converted into embeddings and stored in a vector-capable database. A user question is also embedded. The system searches for document vectors that are similar to the question vector and provides the retrieved text to a foundation model as context.

RAG performance depends heavily on document preparation. The process begins with parsing, which extracts usable text and structure from documents.

The next decision is chunking. Documents are divided into smaller sections that can be embedded and retrieved.

Metadata adds fields such as source, author, date, topic, document type, and access level. It supports filtering, citations, traceability, and permissions.

Protection is also part of preparation. Sensitive data may need to be masked, redacted, or anonymized before indexing.

To conclude Part 2, transformation converts collected data into reliable inputs for machine learning and AI systems. Thank you for watching.